Evolone Layne

evolone@uw.edu
Google Scholar | GitHub | LinkedIn | Portfolio

EDUCATION

University of Washington (*Seattle*, WA)
Ph.D. in Computer Science & Engineering (Robotics) ~Jun 2030
GPA: 3.90 / 4.0

Carnegie Mellon University (*Pittsburgh*, PA)
Master of Human-Computer Interaction (School of Computer Science) May 2025
GPA: 3.96 / 4.0

Howard University (*Washington*, DC)
B.S. in Computer Science in College of Engineering and Computer Science Dec 2023
GPA: 3.79 / 4.0

PUBLICATIONS & PRESENTATIONS

1. **Layne, E**. & Cakmak, M. (2026). " Humanoid Robots in the Wild: A Digital Ethnography via Social Media Videos." *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*.
2. **Layne, E**. & Cakmak, M. (2026). "Humanoid Robots in the Wild: A Digital Ethnography of Online Videos." Special Session on "HRI and Consumer Robots: Building Robots that are Useful and Acceptable" (Sponsored by Meta), ACM/IEEE International Conference on Human-Robot Interaction 2026.
3. Radka, B**.**, **Layne E.,** & Cakmak, M. (2025). "Attitudes towards Humanoid Robots for In-Home Assistance." *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*.
4. **Layne E.** and Mostow, J. “Replicating Bugs Faster”. *Working Papers Journal 9,* pp. 155-156 (2021).

RESEARCH EXPERIENCE

University of Washington
Human-Centered Robotics Lab (Advisor: Maya Cakmak) Sep 2025 – Present

- Executed a user study at PROVAIL using the Hello Robot Stretch RE3 platform; following insights that error tolerance was higher for play tasks, had participants play golf with the robot and conducted in-depth contextual inquiry interviews at each participant's home.
- Conducted two surveys ($n = 40$ each) via Prolific to investigate the privacy-utility trade-offs users expect with in-home humanoid robots, quantifying how desired privacy levels directly impact perceived functionality.

Massachusetts Institute of Technology Lincoln Laboratory
Group 103 – Tactical Autonomy - *Full Secret Clearance* Jul 2026 – Present

- Designed and implemented a replay visualizer for after-action explanation powered by LiteLLM, featuring an integrated chat assistant with context across radio, chat replay, and spatial view widgets, with scenarios generated by Synthex.

Group 42 – Transportation Safety and Resilience - *Full Secret Clearance* Jun 2025 – July 2026

- Designed and implemented interactive data visualization dashboards that transform simulation datasets into interactive, user-friendly visualization, facilitating analysis of agent-based model simulations.

National Aeronautical Space Administration
Ames Research Center**,** Earth-Independent Operations Jan 2025 – July 2025

- Led technical design efforts to develop autonomous problem-solving processes for Mars-bound crew members, addressing communication delays while balancing vehicle constraints and UI feasibility requirements.

Carnegie Mellon University
eHeart Lab (PI: Geoff Kaufman) Aug 2024 – Dec 2024

- Conducted user research to support ESL learners and facilitators in a VR-based environment, through co-design.

RoboTutor (Advisor: Jack Mostow) Jun 2021 – Aug 2021

- Developed a bug replication process with screen recording and JSON-based error tracking

INTERNSHIP EXPERIENCE

Apple Inc

Watch Complications (*Software Engineering Intern*) May 2024 – Aug 2024

- Implemented a watch face using Objective-C with the MVVM architecture, utilizing Swift as the data source
- Added regional time display, gradient backgrounds, color themes, and numeral styles for enhanced personalization

Health Sensing Experience (*Software Engineering Intern*) May 2023 – Aug 2023

- Designed and implemented multiple UI iterations for cycle deviations, using Swift and SwiftUI to enhance usability and support informed patient-doctor discussions
- Analyzed and worked with pre-existing cycle tracking data to determine the most relevant and non-alarming information for display, ensuring a balanced, user-friendly experience

Watch Frameworks (*Software Engineering Intern*) May 2022 – Aug 2022

- Partnered with the Human Interface team to design an accessible watch experience for color-blind users, ensuring inclusivity and usability
- Developed multiple SwiftUI iterations and independently drafted an API proposal to enhance framework functionality and design flexibility

National Aeronautical Space Administration

Jet Propulsion Laboratory (*Reference Mission Systems Engineering Intern*) Jun 2021 – Aug 2021

- Created and documented a step-by-step guide using Jupyter Notebooks to help employees without Linux or terminal experience navigate and resolve merge conflicts effectively

TEACHING & MENTORSHIP

Paul G. Allen School of CSE, Teaching Assistant, CSE 564 Sep 2026 – Dec 2026
Paul G. Allen School of CSE, Research Mentor, CSE 494 Jan 2026 – June 2026
Google Developer Student Club, President Jul 2023 – Aug 2023

AWARDS & FELLOWSHIPS

Graduate

UW Amazon Robotics Fellowship	2026
Black in Robotics Student Conference Travel Grant	2026
Herbold Fellowship	2025
University of Washington College of Engineering Dean's Fellowship	2025
National GEM Consortium Fellowship (MIT Lincoln Lab)	2025
National GEM Consortium Fellowship (Apple)	2024
Carnegie Mellon University Robotics Pathways Fellowship	2024

Undergraduate

Thurgood Marshall College Fund Apple Scholarship	2023
Crown Castle Opportunity Scholarship	2023
Howard University Capstone Scholarship	2020

COMMUNITY SERVICE

Panelist: Google CS Education Week (2023)
Volunteer: Provail (2026)

SELECTED MEDIA COVERAGE

Grow with Google, Watch what we do: Evolone Layne | Grow with Google	2023
Google, Finding the Leader Inside: Evolone's Story as a GDSC Lead at Howard University	2023
Glamour, Evolone Layne Is Taking Over STEM, One App at a Time	2022
BET, Evolone Layne Is Taking Over STEM, One App at a Time	2022
Hilltop, Howard Student Named Glamour's College Women of the Year	2022